Case 08: Deposited Firebrand Consumption

1 Verification purpose

This case verifies removal of deposited firebrands by the consumption model. Paired consumption-off and consumption-on runs use identical forcing, seed, source model, and resolution so their active deposited populations can be compared directly.

The test exposes non-decaying stock, negative active counts, resolution-dependent consumption, and unintended feedback on surface arrival. Success requires the specified population reduction, invariant surface arrival, and convergence toward the finest-grid spatial profile.

2 Mathematical formulation and numerical implementation

Let N'' denote active deposited firebrands per unit area. With mean particle mass $m_{fb} = 0.2$ g, the pile mass loading is

$$\psi = \frac{m_{fb} N''}{10^4} \quad [\mathrm{g\,cm^{-2}}]. \tag{1}$$

The source estimates the local air velocity from the 20-ft wind using a fuel- and canopy-dependent coefficient. It then evaluates the empirical pile heat- flux parameters

$$t_{rise} = \max(-1.02 v_a^2 + 9.08 v_a + 35, 0), \tag{2}$$

$$HF_{rise} = \max(-0.32 v_a^2 + 1.59 v_a + 0.1, 0)\tanh(12\psi), \tag{3}$$

$$HF_{decay} = \max(10^{-6}, -0.05 v_a^2 + 0.02 v_a - 0.01)\tanh(12\psi). \tag{4}$$

After applying the source's heat-flux scaling, the lifetime is the duration for which the modeled incident heat flux exceeds $HF_{fb,live} = 10$ kW m^{-2}:

$$t_{fb,live} = \max(t_{half,2} - t_{half,1}, 0). \tag{5}$$

The intended explicit update is

$$N''^{n+1} = N''^{n} - \frac{N''^{n}}{\max(t_{fb,live}, t_{min})}\Delta t, \qquad t_{min} = 10\ \mathrm{s}. \tag{6}$$

ELMFIRE stores pixel counts, so the implementation multiplies the areal decay rate by pixel area before subtracting from accumulated firebrand deposition. Only the consumption-on member calls this consumption calculation. Final accumulated firebrand deposition therefore represents the active deposited population. The transient time-resolved firebrand deposition product records new deposition during an output interval and is reset after each dump; it is not used as an active-stock surrogate. Instead, independent consumption-on runs terminated at predeclared times provide the temporal samples from their final active-stock rasters.

The current source is also audited as part of this test. It presently assigns a minimum firebrand lifetime of 1E9 s, whereas the documented model requires 10 s [1], as expressed in Equation 6. The verification preserves this behavior and is expected to fail the consumption-effect metric until that implementation discrepancy is corrected.

3 Derivation of expected behavior

For otherwise identical runs, let $A_{on}(\Delta x)$ and $A_{off}(\Delta x)$ be the centerline sums of active deposited particles at 260 s. The retained fraction is

$$f_{ret}(\Delta x) = \frac{A_{on}(\Delta x)}{A_{off}(\Delta x)}. \tag{7}$$

The documented result that more than 96% is consumed gives $f_{ret} \leq 0.04$ [1]. Because the delayed-ignition baseline uses long ignition delays and is surface-fire dominated, removing deposited embers should not materially change time of arrival (TOA). The active-particle line density is recovered from a centerline pixel count as $n'(x_i) = N_i/\Delta x$ in pcs/m.

For grid convergence, the 5, 10, and 30 m consumption-on profiles are linearly interpolated to the 1 m cell centers. Their normalized integral error is

$$E_{\Delta x} = \frac{\int |n'_{\Delta x}(x) - n'_1(x)|\, dx}{\int |n'_1(x)|\, dx}. \tag{8}$$

Consistent refinement requires $E_{30} \geq E_{10} \geq E_5$, with $E_5 \leq 0.15$.

For a cell whose calculated lifetime is $T_n = \max(t_{fb,live,n}, 10)$, Equation 6 gives the explicit retained fraction

$$\frac{N''_{n+1}}{N''_n} = 1 - \frac{\Delta t_n}{T_n}, \qquad \frac{N''_m}{N''_0} = \prod_{n=0}^{m-1} \left(1 - \frac{\Delta t_n}{T_n}\right).$$

This is the reference update that the consumption-on run must execute; the consumption-off reference has every factor equal to one. The published statement that more than 96 percent is consumed is converted without further fitting to

$$1 - f_{\text{ret}} \geq 0.96 \quad \Longleftrightarrow \quad f_{\text{ret}} = \frac{A_{\text{on}}}{A_{\text{off}}} \leq 0.04.$$

The nominal surface Courant–Friedrichs–Lewy (CFL) upper bounds are $0.5\Delta x/0.71$. To keep every requested 10-s-multiple observation time exact, preprocessing selects the largest divisor no greater than that bound, preferring whole seconds when feasible:

$$\Delta t = \{0.666667,\ 2,\ 5,\ 10\}\ \text{s}$$

for $\Delta x = \{1, 5, 10, 30\}$ m. Thus all 260-s paired stops and 50–500-s history stops are exact timestep multiples, and every actual surface CFL remains below 0.5. The intended lifetime is recalculated locally from v_a and $\psi = 0.2N''/10^4$, so no constant lifetime is invented for the report.

For the paired control, the exact expected TOA difference is zero and the active-stock lower bound is zero particles. The convergence reference is the generated 1-m profile: $E_1 = 0$, with expected refinement ordering $E_{30} \geq E_{10} \geq E_5 \geq E_1$. The decision substitutes $E_5 \leq 0.15$, paired TOA difference ≤ 10 s, and minimum active count $\geq -10^{-6}$, in addition to the 0.04 retained-fraction limit.

4 Verification metrics and acceptance criteria

All 48 prescribed simulation conditions are required: eight paired 260-s metric runs and 40 consumption-on stop-time runs (four resolutions at ten sample times). Exactly one final active-flux and TOA raster per simulation condition is required. The two-cell numerical halo is excluded. The components are:

1. $\max_{\Delta x} f_{ret} \leq 0.04$;
2. paired centerline TOA difference ≤ 10 s at every resolution;
3. every active count $\geq -10^{-6}$ particles;
4. $E_5 \leq 0.15$; and
5. $E_{30} \geq E_{10} \geq E_5$.

The overall decision is the conjunction of completeness and all five components. Missing, duplicated, dimensionally stale, or unreadable outputs produce an insufficient-output result rather than a pass.

4.1 Justification of metric quantification

The retained-fraction ceiling of 0.04 is not tuned to these outputs; it is the complement of the reported result that more than 96% of deposited firebrands are consumed by the evaluation time [1]. The paired TOA allowance of 10 s is a mechanism-isolation check at the scale of one output-time increment: consumption may alter the active stock but must not materially change the surface-dominated arrival field. The lower bound of -10^{-6} particle is only a floating-point allowance around the exact nonnegativity invariant and cannot conceal a physically meaningful negative population.

The convergence ordering is more informative than agreement at one grid. It requires every refinement to move toward the 1 m profile. The $E_5 \leq 0.15$ limit allows cell averaging of a steep transported-stock profile while requiring at least 85% agreement in normalized integral content; errors of that size remain clearly separated from the coarse-grid shape changes the sweep is intended to expose. All paired and stop-time runs are mandatory because omitting one would break either the on/off control or the temporal consumption history.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Eight paired metric runs plus 40 active-stock stop-time runs, with final-raster shape/nodata validation and halo exclusion.	48 of 48 simulation conditions
Retained active fraction	For each resolution, consumption-on active population divided by its consumption-off pair at 260 s.	Maximum ≤ 0.04
Surface-TOA invariance	Maximum paired centreline TOA difference on common finite physical cells.	≤ 10 s at every resolution
Active-count nonnegativity	Minimum value over every active-firebrand raster.	$\geq -10^{-6}$ particles
Fine-grid profile convergence	Normalized L_1 error of the 5-m consumption-on profile relative to the 1-m profile.	$E_5 \leq 0.15$
Ordered convergence	Normalized profile errors for 5, 10, and 30-m grids.	$E_{30} \geq E_{10} \geq E_5$
Overall decision	Conjunction of completeness and all five physical/numerical checks.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for the 10 m grid with firebrand consumption enabled. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

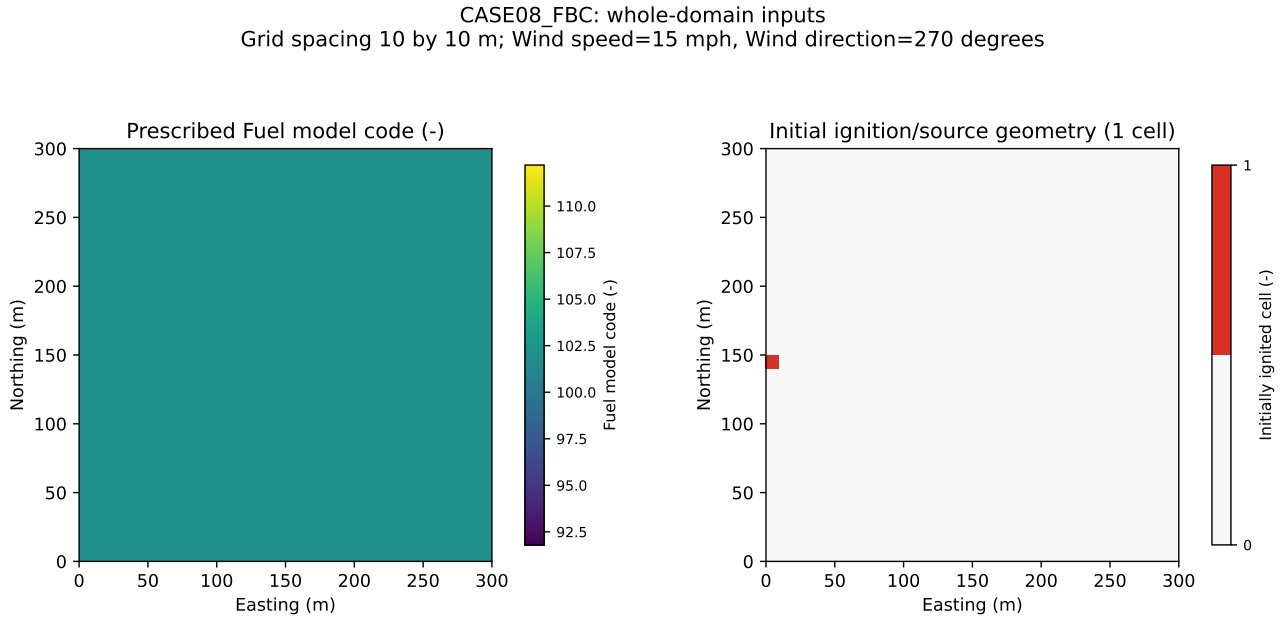


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Parameter	Configured value	Role
Physical domain	300×300 m	identical extent at all resolutions
Cell sizes	1, 5, 10, 30 m	active-stock convergence sweep
Numerical halo	2 cells per side	excluded from all metrics
Metric evaluation time	260 s	paired consumption-on/off comparison
History sample times	50–500 s in 50-s increments	final active-stock samples for the temporal curve
Temporal CFL	at most 0.5 based on 0.71 m/s	conservative exact-duration divisors
Wind	15 mph from 270 degrees	uniform downwind forcing
Generation	PER-MW, $33.3/\Delta y$ pcs/s/MW	delayed-ignition baseline vegetation source
Transport/accumulation	EMPIRICAL/EULERIAN	active-stock implementation
Ignition	PHYSICAL; 300 s development	prevents spotting-dominated spread
Consumption	paired off/on	isolates finite lifetime
Critical live heat flux	10 kW/m^2	hard-coded lifetime threshold
Crosswind distribution	disabled	one-dimensional profile diagnostic

Table 2: Configuration used by every paired simulation condition.

Before inspecting results, consumption-on curves should be far below the paired no-consumption

curve upstream of 200 m, retained fractions should lie below the 4% line, and progressively finer active-stock profiles should agree. TOA should be nearly unchanged between paired runs.

6 Actual simulation results

Figure 2 shows the representative accumulated firebrand density from the 10 m grid with firebrand consumption enabled. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. Counts are divided by physical cell area, giving pcs m^{-2}, so the retained and consumed portions of the deposit field can be interpreted spatially.

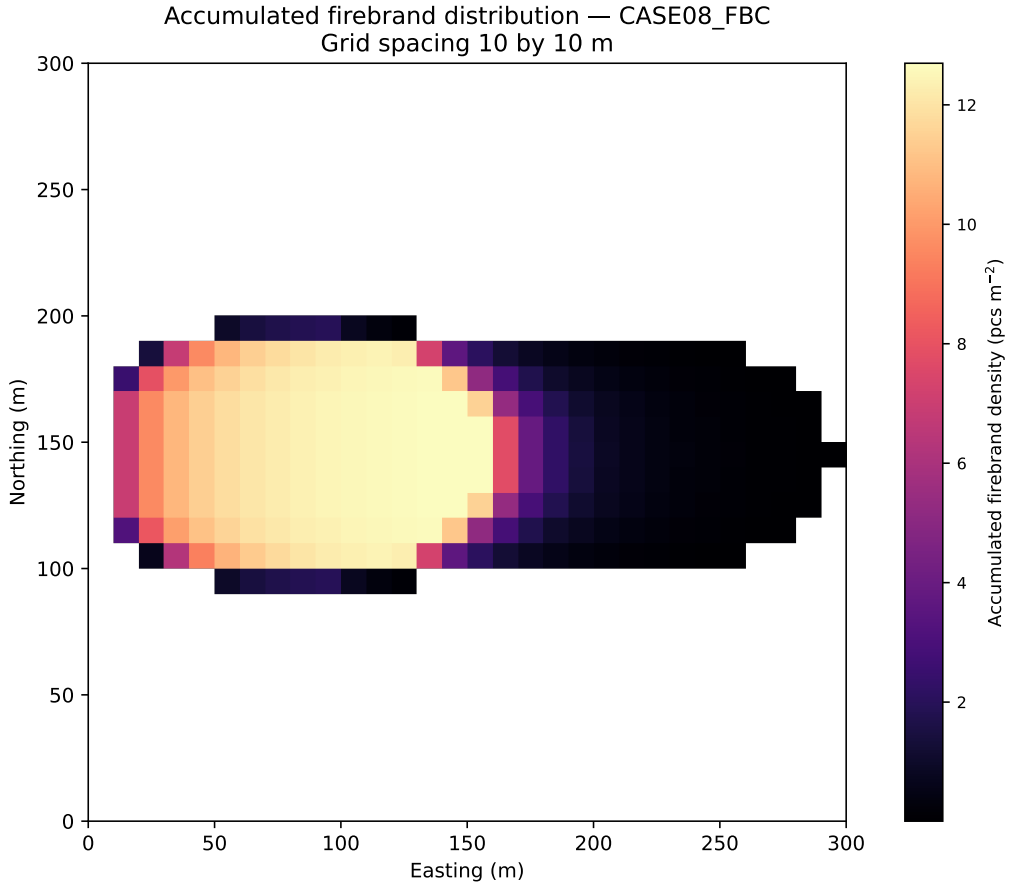


Figure 2: Whole-domain accumulated firebrand density from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

All 48 simulation conditions were rerun. The temporal curves use ten independent consumption-on stop-time runs per resolution, from 50 to 500 s in 50 s increments. Each plotted sample is therefore a final ELMFIRE active-stock field, not an interval-deposition raster. Figure 3 first presents the four physical observables for the finest consumption-on simulation condition: active firebrands, arrival time, leading-edge position, and leading-edge rate of spread (ROS).

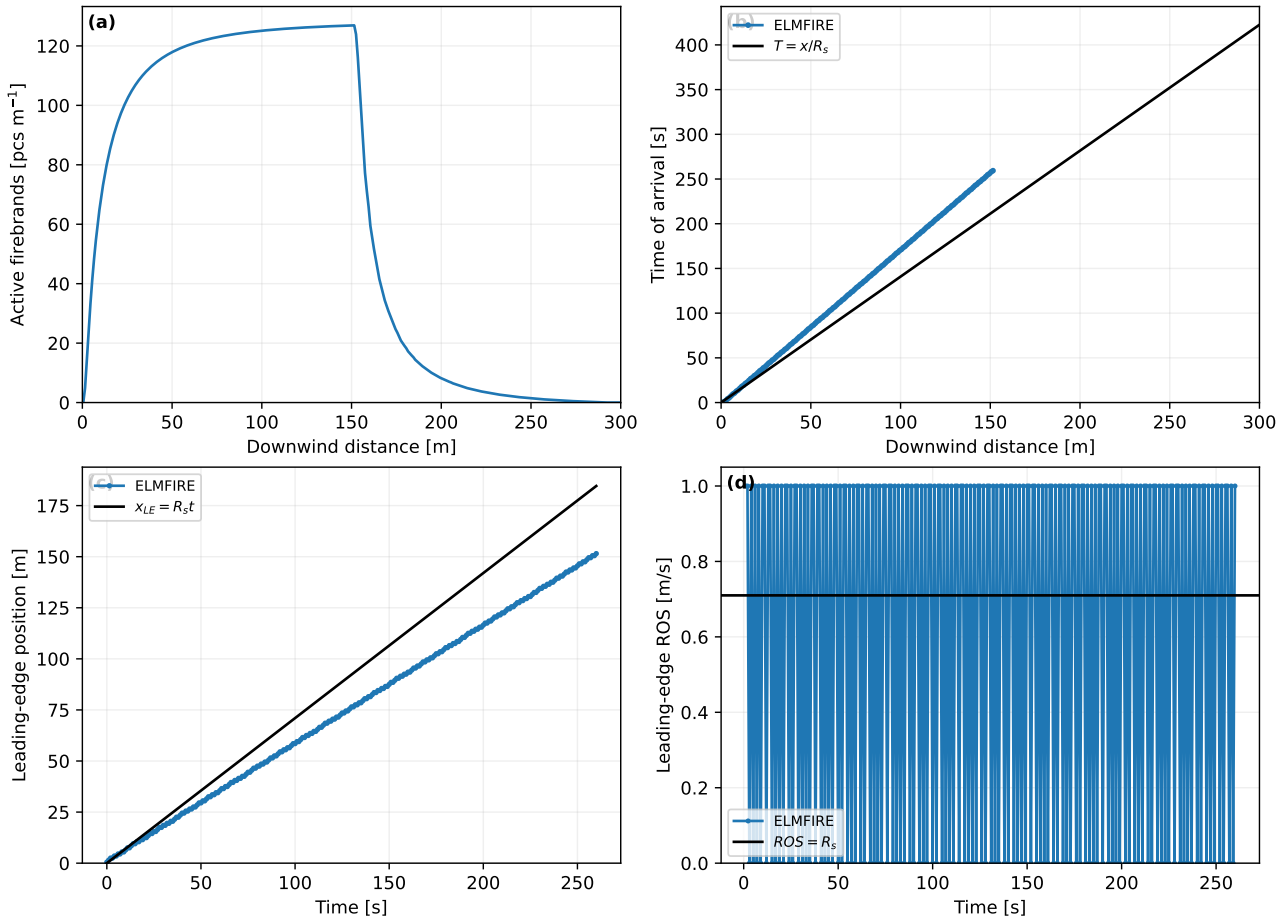


Figure 3: Consumption-on state at 1 m resolution. The top row gives the active firebrand and arrival-time profiles; the bottom row gives the reconstructed leading-edge trajectory and ROS with surface-spread references.

Figure 4 reproduces the temporal and spatial resolution views of the active population. The left panel samples the physical location nearest 200 m; the right panel selects the dump nearest 250 s.

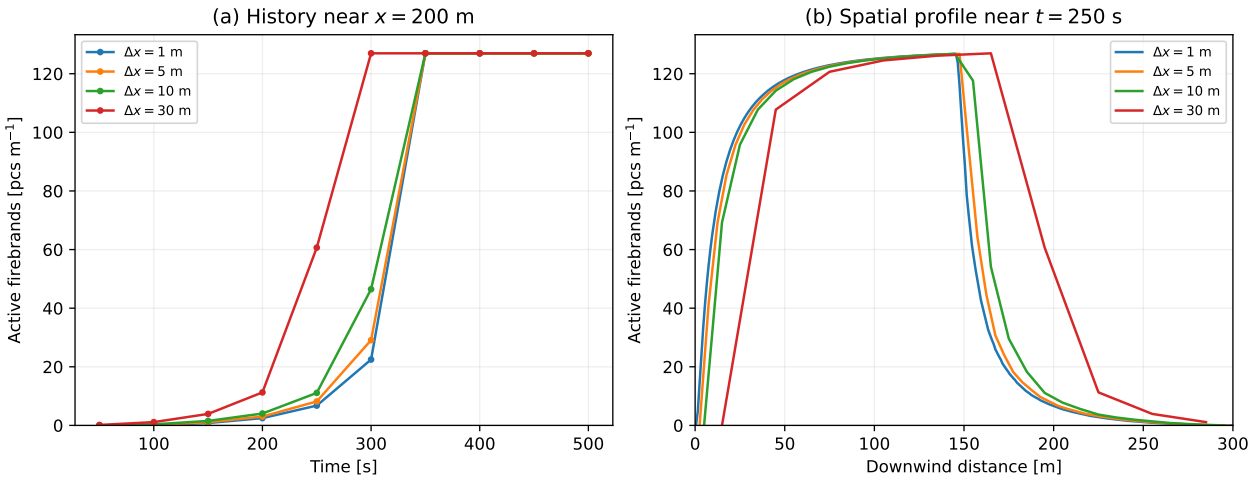


Figure 4: Consumption-model resolution sweep. Left: active-firebrand history near 200 m. Right: active-firebrand spatial profile near 250 s. Every curve is read from a predeclared stop-time run's final ELMFIRE raster.

Figure 5 is the supplemental paired diagnostic that makes the consumption-on/off effect and the 4% retention criterion explicit.

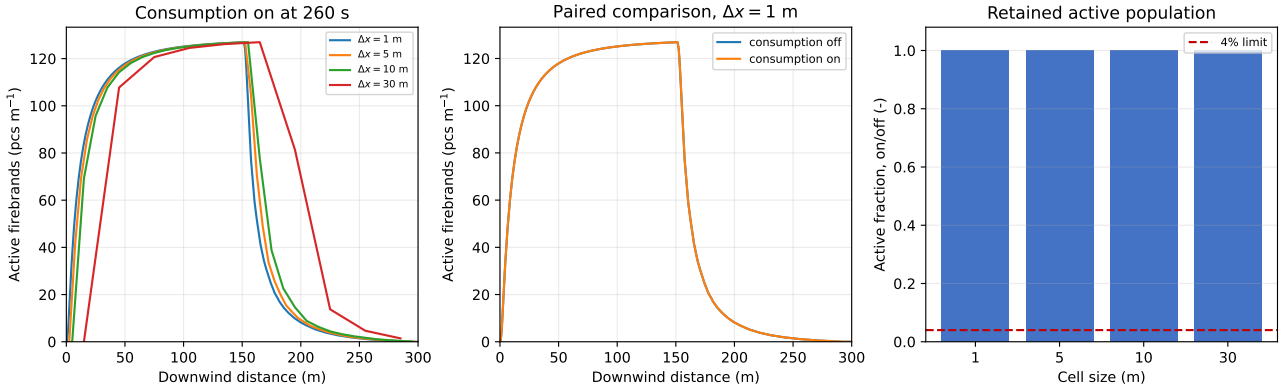


Figure 5: Paired consumption diagnostic. Left: consumption-on profiles by resolution. Center: paired 1 m profiles. Right: retained active fraction with the 4% acceptance limit.

7 Calculated metrics and verification decision

8 Technical interpretation

All four measured retained fractions are exactly 1.0, so enabling consumption does not change the active deposited population. The maximum paired TOA difference is 0 s, while the normalized profile errors are 0.0363, 0.0778, and 0.3511 at 5, 10, and 30 m, respectively. The source-normalized profiles therefore converge monotonically and the paired dynamics remain controlled, but the primary consumption metric fails decisively. Given the audited 10^9 s minimum lifetime, this is a source-code implementation defect rather than insufficient run duration. A TOA-only result cannot establish consumption behavior, which is why TOA appears here solely as a controlled invariance check.

Metric	Limit	Calculated	Status
Output completeness	48 simulation conditions	48/48	–
Maximum retained fraction	≤ 0.04	1	false
Maximum paired TOA difference	≤ 10 s	0	true
Minimum active count	$\geq -10^{-6}$	0	true
5 m normalized profile error	≤ 0.15	0.036343	true
Ordered profile convergence	required	–	true
Overall decision	all pass	false	**FAIL**

Table 3: Calculated verification metrics for the completed simulations.

9 Scope and limitations

This test verifies active-stock decay in one flat, uniform delayed-ignition baseline-like configuration. It does not validate the empirical heat-flux correlation, pile clustering, subgrid redistribution, or the 10 s lifetime against experiments. The 50 s sequence of independently stopped runs provides the active-stock history near 200 m and the spatial profile near 250 s directly from ELMFIRE rasters. The experiment still does not resolve within-cell firebrand clustering or validate the lifetime correlation against measurements.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.